VA DIVISION A • GRADES 3–4

Bungee Egg Drop

Official Study Guide • 2025–2026 Season

Official Rules

Scoring Formula

Conservation of Energy

Gravitational PE

Elastic PE

Kinetic Energy

Event Overview & Official Rules

Rules

Design a bungee cord system using rubber bands to protect a raw egg — dropping from a two-story stairwell and stopping as close as possible to the Artificial Horizon WITHOUT going past it.

MATERIALS PROVIDED

- Assortment of rubber bands: sizes #64, #33, #19, and #117B
- 50 cm of 2.54 cm masking tape
- 1 large raw egg (choose from those provided)
- 1 sandwich or snack-sized plastic bag
- The egg must be sealed inside the plastic bag

BUILD RULES (25 min)

- Build + written test happen simultaneously (25 minutes)
- Rubber bands may be cut or left whole — tie in any manner
- No tape may support or strengthen the egg or the plastic bag
- Tape only used to attach bungee cord to the bag
- Estimate your best drop distance — announced to supervisor

SAFETY REQUIRED

- ANSI Z87 D3 Splash/Droplet standard goggles required
- Must be worn during ALL event testing — all competitors
- Eyeglasses are NOT safety rated
- Without proper goggles = no-show score
- This is a D3 (splash) rating — NOT just impact glasses

Scoring Formula — Official Calculation

Scoring

Final Score = Assessment Score + the BETTER of your two Drop Scores. Max = 100 pts. The goal: stop the egg as close to the Artificial Horizon as possible WITHOUT going below it.

Term	Formula / Definition	Notes
Final Score (FS)	FS = AS + max(DS1, DS2)	Take your BEST drop score
Assessment Score (AS)	AS = 50 × (your RAS ÷ best RAS among all teams)	Max 50 pts
Drop Score 1 (DS1)	DS1 = 50 × (TDD − DD1) ÷ max(TDD − DD for all teams)	Closer = higher score
Drop Score 2 (DS2)	DS2 = 50 × (TDD − DD2) ÷ max(TDD − DD for all teams)	Same formula, 2nd drop
Target Drop Distance (TDD)	Announced distance from railing to Artificial Horizon. Min = 200cm	Set by Event Supervisor
Drop Distance (DD)	200cm if egg passes below horizon; else distance from horizon to lowest egg point	Measured from video
Egg Broken?	Egg rolled on paper towel — ANY wet spot = broken. Cracked but not leaking = OK.	Broken = Tier 2
Tiebreaker 1	Smallest difference between your ESTIMATED and ACTUAL drop distance	Estimate accuracy
Tiebreaker 2	Correct answers on selected written assessment questions	Assessment quality

Example: TDD=400cm. Team RAS=6, best=9 → AS=33.33. 1st drop below horizon → DS1=25.45. 2nd drop 13cm → DS2=49.24. FS=33.33+49.24=82.57

📊 The Artificial Horizon & Competition Phase

Competition

The Artificial Horizon is a chart marked from −100cm to +100cm with zero in the middle. Your goal: stop the egg as close to ZERO as possible — without going past (below) the zero line.

1 Attach Bungee to Retaining Hook

Attach the end of your bungee cord OPPOSITE the egg to the Retaining Hook on the Bungee Jump Platform. Rest of rubber bands hang loose over the front of the platform.

2 Hold Egg at Platform Edge

Place and hold the egg at the end of the jump platform in a VERTICAL position. Do not lean, swing, or push.

3 Countdown & Release

Announce 'ready' to the Event Supervisor. Wait for reply that camera is ready. Count down from 3 (3, 2, 1, drop) and OPEN YOUR HAND — let the egg fall. DO NOT PUSH THE EGG.

4 Slow-Motion Video Recorded

Supervisor records slow-motion video showing the ENTIRE Artificial Horizon Chart. Distance from horizon to lowest egg point is measured to the nearest 1cm mark.

5 Two Drops Allowed

You receive TWO drops. Between drops, you may adjust your bungee cord. Your BEST drop score is used — not both averaged.

6 Egg Inspection

Egg is removed from the plastic bag and rolled on a paper towel. Any WET area = broken (Tier violation). Cracks without leakage = NOT

⚠ Tier Violations — Know What Costs You Points

Tiers

✅ TIER 1 Full Score	The team device meets ALL building requirements. The egg remains unbroken (no leakage). The team follows the Spirit of the Problem guidelines. **Penalty: No penalty — full score calculation applies**	FULL SCORE
⚠ TIER 2 Score × 0.80 (−20%)	The team device is deemed to have a construction OR timing fault violation, AND the team otherwise follows the Spirit of the Problem guidelines. **Penalty: 20% deduction from final score**	80% SCORE
❌ TIER 3 Score × 0.70 (−30%)	The team did not follow the Spirit of the Problem guidelines — this means attempting to gain an unfair advantage through rule interpretation or other means. **Penalty: 30% deduction from final score**	70% SCORE

⚡ Law of Conservation of Energy

Energy Law

The written assessment focuses on: Law of Conservation of Energy, Gravitational Potential Energy (GPE), Kinetic Energy (KE), and Elastic Potential Energy (EPE).

LAW OF CONSERVATION OF ENERGY

- ◆ Energy cannot be created or destroyed — only transformed from one form to another
- ◆ Total energy of a closed system remains constant throughout the bungee jump
- ◆ At the top (rest): ALL energy is Gravitational Potential Energy (GPE)
- ◆ At lowest point (momentarily stopped): ALL energy is Elastic PE (stored in stretched bands)
- ◆ During fall: GPE converts to KE. As bands stretch: KE converts to Elastic PE

ENERGY TRANSFORMATIONS

- ◆ 1. Start (rest at top): GPE = mgh, KE = 0, Elastic PE = 0
- ◆ 2. Falling freely: GPE decreasing → KE increasing
- ◆ 3. Bands start stretching: KE decreasing → Elastic PE increasing
- ◆ 4. Lowest point: KE = 0, GPE minimum, Elastic PE maximum
- ◆ 5. Rebound: Elastic PE → KE → GPE again (cycle repeats, loses energy to heat/sound)

IDEAL vs. REAL

- ◆ In ideal system: total energy is perfectly conserved (no losses)
- ◆ In real system: some energy lost as heat (rubber warms up) and sound
- ◆ This is why the egg doesn't bounce back to EXACTLY the starting height
- ◆ Rubber bands are not perfectly elastic — they have hysteresis (energy loss)
- ◆ Your goal: use this energy loss to your advantage — egg stops near horizon, doesn't snap back past it

Energy Formulas — GPE, KE, Elastic PE

Formulas

GPE

Gravitational Potential Energy

$GPE = m \times g \times h$

m = mass (kg) • g = 9.8 m/s² • h = height above reference point (m)

Key: GPE increases with HEIGHT. Higher drop = more GPE = more energy to convert. The egg has maximum GPE at the start of the drop.

📌 *Egg mass = 0.06 kg, drop = 3m: GPE = 0.06 × 9.8 × 3 = 1.76 Joules*

KE

Kinetic Energy

$KE = \frac{1}{2} \times m \times v^2$

m = mass (kg) • v = velocity (m/s)

Key: KE is ZERO at the start (no motion) and maximum just before the bands start stretching. Velocity doubles → KE quadruples.

📌 *Egg at 4 m/s: KE = 0.5 × 0.06 × 16 = 0.48 J. At 8 m/s: KE = 1.92 J (4× more)*

EPE

Elastic Potential Energy

$EPE = \frac{1}{2} \times k \times x^2$

k = spring constant (stiffness of bands) • x = stretch distance (m)

Key: EPE stored in stretched rubber bands. More stretch = more EPE stored. EPE is ZERO until bands become taut, then increases rapidly.

📌 *Stiffer bands (higher k) store more energy per cm of stretch. Adding bands in parallel increases k.*

Bungee Cord Design Strategy

Design

Goal: design a rubber band bungee that extends the egg to EXACTLY the Artificial Horizon — no further. Think of it as an elastic spring: the more bands, the stiffer; longer bands, the more stretch.

RUBBER BAND BASICS

- #19 = shortest/stiffest • #33 = medium • #64 = longer • #117B = longest
- Bands in SERIES (end-to-end): longer bungee, more stretch per force
- Bands in PARALLEL (side-by-side): stiffer bungee, less stretch per force
- Mix sizes: short stiff bands stop the egg; long bands provide initial give
- You may CUT bands to any size — a cut band is just a shorter band

ESTIMATING STRETCH

- Test on the floor: attach your bungee to a fixed point, hang the bagged egg
- Measure: how far does it stretch before stopping the egg?
- Compare to TDD (announced at start) — adjust number/size of bands
- Add bands to make bungee stiffer (stops sooner). Remove to let it go further.
- The egg is attached at the BOTTOM — bungee length + stretch = total travel

ADJUSTMENTS BETWEEN DROPS

- You get TWO drops — use the first drop as data, adjust for drop 2
- Went past horizon? Remove bands or use shorter/stiffer configuration
- Stopped short? Add longer bands or remove shorter ones
- Small adjustment: cut one band shorter. Large adjustment: add/remove entire band
- Write down your estimate accurately — tiebreaker is closest estimate to actual

Written Assessment — What to Study

Assessment

The written assessment tests Conservation of Energy, GPE, KE, and Elastic PE. It occurs SIMULTANEOUSLY with the 25-minute build phase. You must answer questions while building.

CONSERVATION OF ENERGY

1. What is the Law of Conservation of Energy?
2. Where is GPE maximum in the bungee drop?
3. Where is KE maximum?
4. Where is EPE maximum?
5. Why doesn't the egg return to its exact starting height?

ENERGY FORMULAS

1. GPE = m × g × h: identify what each variable means
2. KE = ½mv^2: what happens when velocity doubles? (KE × 4)
3. EPE = ½kx^2: what is k? What is x?
4. If egg falls 4m, calculate GPE (given mass and g = 9.8)
5. At the lowest point, what has all the GPE become?

BUNGEE SYSTEM DESIGN

1. Why does adding more rubber bands change the drop distance?
2. What is the difference between bands in series vs. parallel?
3. How does the mass of the egg affect the drop distance?
4. What type of energy is stored in a stretched rubber band?
5. Why is the egg placed in a plastic bag?

📖 Key Vocabulary — Know These Cold

Vocab

Conservation of Energy:	Energy cannot be created or destroyed — only converted between forms
Gravitational PE (GPE):	Energy stored due to height: GPE = mgh. Max at start of drop.
Kinetic Energy (KE):	Energy of motion: KE = ½mv². Max when bands just start to stretch.
Elastic PE (EPE):	Energy stored in a stretched/compressed elastic: EPE = ½kx². Max at lowest point.
Spring Constant (k):	Stiffness of the bungee — higher k = stiffer = less stretch per unit force
Artificial Horizon:	The zero-line on the competition chart — target for the egg to stop at

Target Drop Distance:	Announced distance from the railing to the Artificial Horizon (TDD, min 200cm)
Drop Distance (DD):	Actual distance from horizon to lowest point of egg travel
Bungee in Series:	Bands connected end-to-end: longer, stretches more, softer
Bungee in Parallel:	Bands side-by-side: shorter total, stiffer, less stretch
Hysteresis:	Energy lost as heat/sound when rubber bands stretch — why egg doesn't bounce fully back
Tier Violation:	Penalty for breaking build rules: Tier 2 = −20%, Tier 3 = −30% of final score